

ASBIR KALLADA

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PROFESSIONAL SUMMARY

Recent CSE graduate skilled in Python and machine learning with hands-on IoT and full-stack project experience. Published researcher (IJIRT) and former AI/ML intern, seeking entry-level Python or software developer roles where I can build practical, scalable solutions. Focused on Python development and delivering deployable ML/IoT solutions during internships and projects.

TECHNICAL SKILLS

Programming Python, Java, C, Dart, SQL, OOP, Data structures, Algorithms

Data Science & ML Pandas, NumPy, Matplotlib, Scikit-Learn, EDA

AI/GenAI ChatGPT, OpenAI API, LangChain, Claude AI, Cursor, DeepSeek, Gemini API, Prompt Engineering, Hugging Face, Streamlit

Web Development HTML, CSS, JavaScript, React.js, Node.js, REST API

Tools & Technologies Git, GitHub, MongoDB, Postman, VS Code, Jupyter Notebook, Linux

Embedded & IoT ESP32, Ultrasonic Sensors, Hardware-Software Integration

Soft Skills Problem Solving, Leadership, Communication, Event Organization, Teamwork

Languages English, Malayalam, Hindi, Urdu, Arabic

INTERNSHIP

AI/ML & Data Science Intern

'23 — '24

Capabl India

- Designed and implemented 5 end-to-end data science projects (Spaceship Titanic ML Challenge, Indian Marriage Dataset Predictions, Baseball Player Data Analysis, Interactive Census Dashboard, Weather Dashboard), delivering modular code and reproducible pipelines for review.
- Built stakeholder-facing visualizations and interactive dashboards (HTML/JS) for the Interactive Census Dashboard and Weather Dashboard, integrating external APIs to enable data-driven exploration.
- Applied feature engineering, classification and regression techniques across projects to produce validated models and clear analytical summaries used in internal presentations and decision-making.

PUBLICATIONS

A Survey on AI-Powered Smart Assistive Cane for Visually Impaired Using IoT and Computer Vision

Apr '26

International Journal of Innovative Research in Technology (IJIRT)

IJIRT, April 2026 — Registration ID: 197375. Research focus: AI-powered obstacle detection, Edge AI processing, IoT sensor fusion, and computer vision for assistive mobility.

PROJECTS

NovaCane – Smart AI Cane for Visually Impaired [Link](#)

'25 — '26

- Result: Prototype provided real-time vibration alerts for detected obstacles using an ESP32-WROOM-32 microcontroller.
- Action: Developed obstacle detection logic and integrated ultrasonic sensors with the ESP32-WROOM to trigger haptic feedback.
- Challenge: Needed to validate hardware-software integration and ensure reliable obstacle detection in real-world testing.

SNGCE Workflow Automation System [Link](#)

'24 — '24

- Result: Replaced paper-based college processes with a digital workflow tool adopted by 1800+ students and staff across SNGCE.
- Action: Implemented a centralized digital workflow to remove paper bottlenecks and track requests for students and staff.

Parkinson's Disease Detection [Link](#)

'24 — '24

- Result: Compared performance of four ML models on 195 biomedical voice samples to identify suitable approaches for voice-based detection.
- Action: Trained Logistic Regression, Decision Tree, Random Forest, and SVM models on the dataset to evaluate comparative performance.

IPL Data Analysis & Visualization [Link](#)

'24 — '24

- Result: Produced visualizations that summarize season-over-season trends and highlight notable player and team performance patterns.
- Action: Conducted exploratory data analysis on multi-season IPL datasets to surface performance trends and perfect output obtained.

Shopping Aggregation Platform [Link](#)

'23 — '23

- Result: Enabled streamlined product comparison and improved user experience for multi-site searches.
- Action: Built API-based data fetching, implemented search functionality, and created a responsive UI with cross-browser compatibility.
- Challenge: Needed to aggregate product prices across e-commerce sites to simplify multi-site searches for users.

EDUCATION

B.Tech in CSE (Computer Science and Engineering), Sree Narayana Gurukulam College of Engineering (SNGCE) (GPA: 6.68)

'22 — '26
Ernakulam

Higher Secondary (12th) & SSLC (10th), MUESS

'19 — '22
Kondotty

CERTIFICATIONS

Data Science Certification , Capabl India
Machine Learning Certification, Capabl India
Python Programming Certification , Capabl India

LEADERSHIP & EXTRA-CURRICULAR

Arts Club Secretary, Student Union Council, SNGCE — 2025 - 2026

- Organized and managed college-level Tech Fest, Arts Fest, and Logo Launch events end-to-end.
- Handled vendor budgeting, audio/lighting/staging specifications, and cross-team scheduling with the Student Council.
- Demonstrated leadership coordinating multi-department activities and formal proposals to college authorities.

NSS Volunteer — Participated in community service and awareness activities.